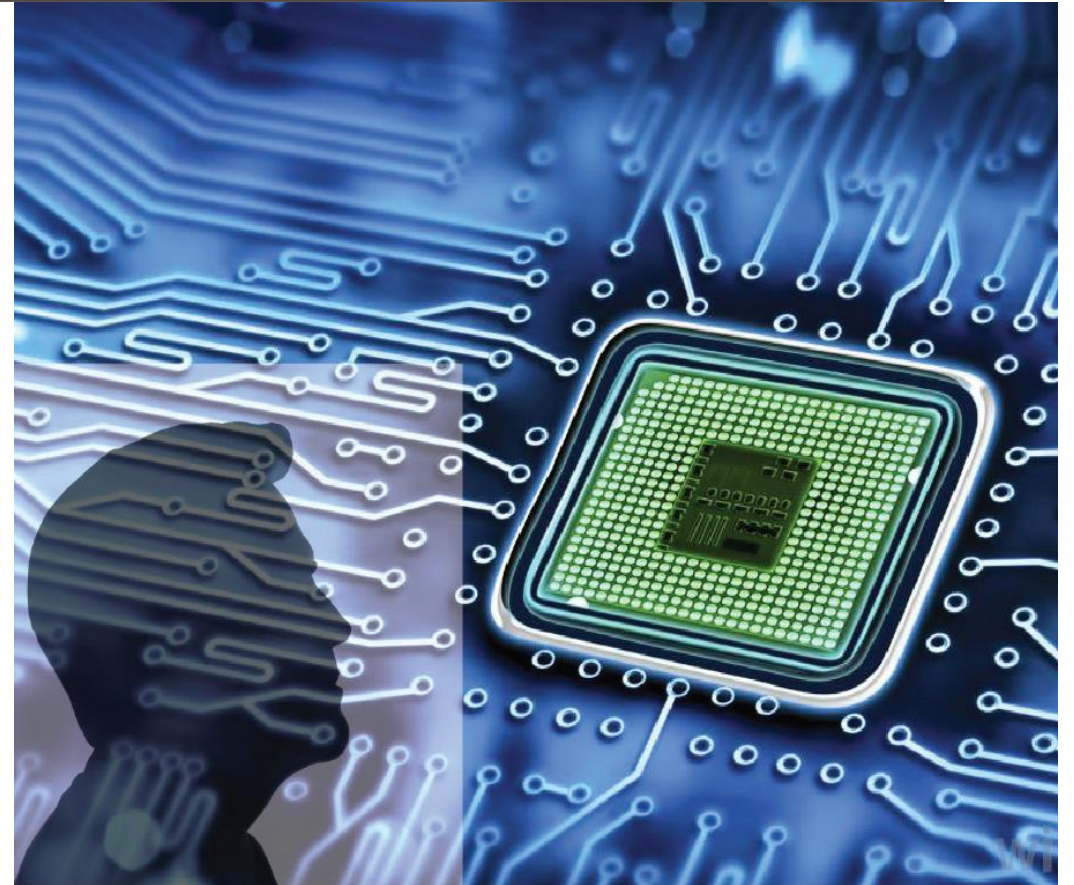


VLSI DESIGN FLOW

Week-1, Lecture-1 Introduction

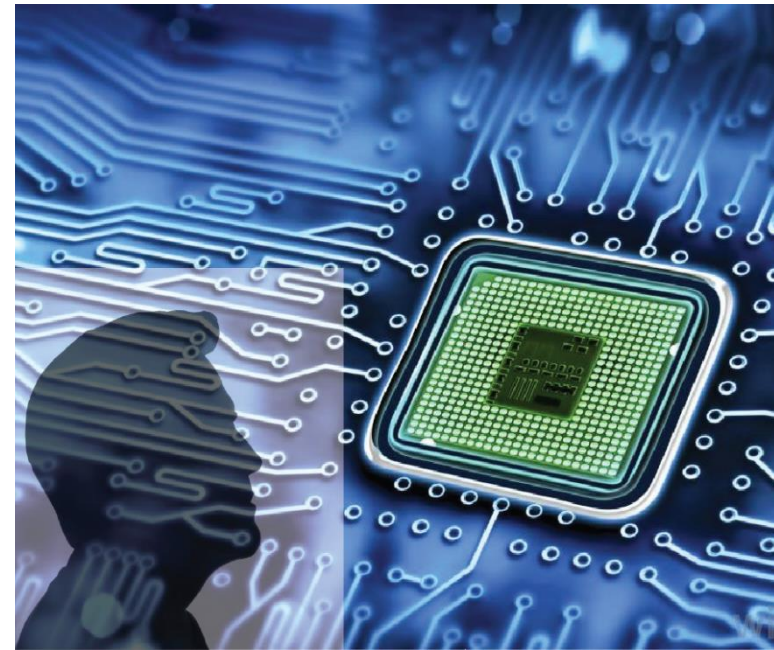
Sneh Saurabh
12th August, 2024



About the Course

VLSI Design Flow (VDF) : Objective

- Develop a basic understanding of the methods, tools and technologies that go into transforming an *“idea”* into an *“integrated circuit”*.
- Employ the-state-of-the-art CAD tools and gain a *practical understanding* of the VLSI design flow.



VDF : Expected outcome

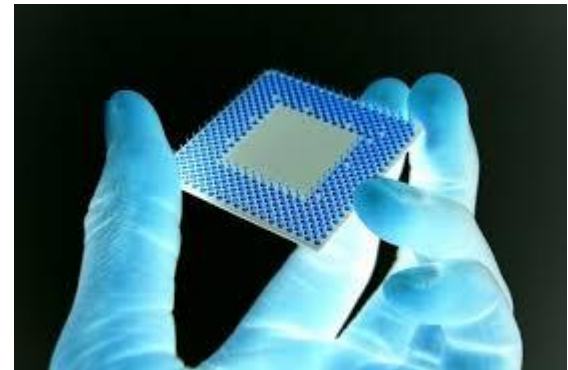
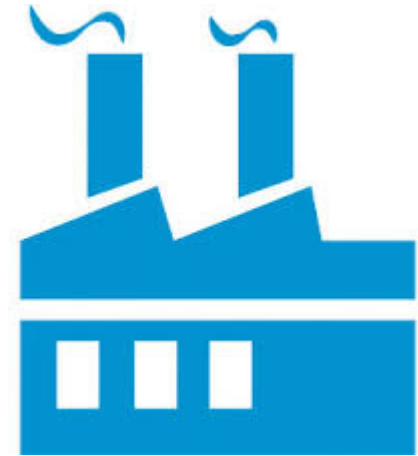
Students are able to

- Explain the **design and verification steps** in the VLSI design flow and their purpose and
- Evaluate various **trade-offs** that need to be made at various steps in the VLSI design flow.
- **Design and verify simple VLSI circuits** using the state-of-the-art computer aided design (CAD) tools at different levels of abstractions.
- Apply representative **algorithms** that are used in implementing CAD tools to design and verify integrated circuits.



VDF : Where to apply the learning?

- Will be used extensively in the semiconductor industry
- Applications in research on VLSI, CAD, Semiconductors, Circuit Design, FPGA, Embedded Systems, Computer Architecture



VDF: Course Content

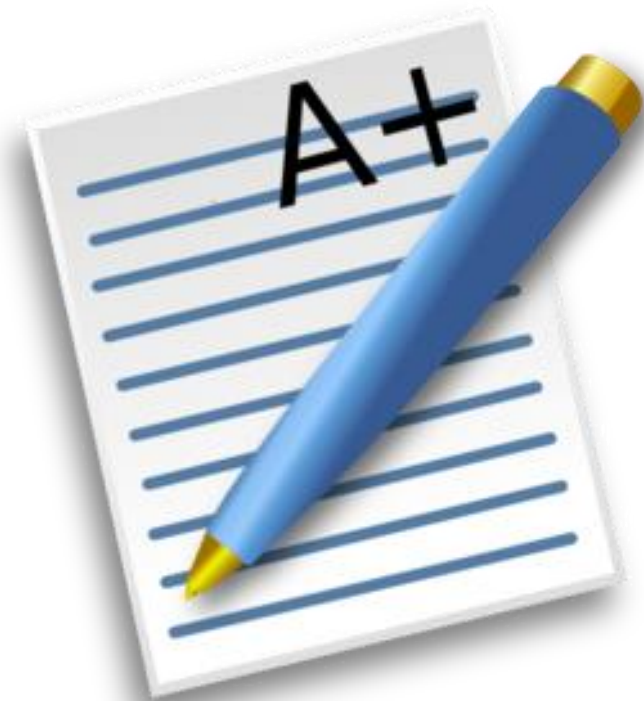
Week No.	Topic
1	Introduction: Basic Concepts; Design Styles; Designing vs. Fabrication; Processes involved in taking "Idea" to RTL: Software/Hardware Partitioning, Behavioral Synthesis
2	Overview of "RTL to GDS" Design Implementation, Verification and Test Flow; Processes after GDS Tapeout to Final Chip;
3	Design capture: HDL (Verilog), RTL Simulation and Synthesis;
4	Introduction to Formal Methods: Equivalence Checking, BDD, SAT;
5	Library and Constraints; Static Timing Analysis; Power Analysis;
6	Logic Synthesis: Optimization, Technology Mapping;
7	Design for Test (DFT): Fault Models, Scan Insertion,
8	ATPG, BIST
9	Basics of Physical Design
10	Partitioning, Floorplanning, Power Planning;
11	Placement: Techniques and Optimization; Clock Tree Synthesis: Distribution, Skew Optimization;
12	Routing: Global and Detailed; Physical Verification: Extraction, LVS, ERC, DRC;
13	ECO, Sign-off

VDF: Google classroom

- Google classroom
 - Announcements
 - Sharing lecture slides, assignments etc.
 - Submission of assignments
 - Link: <https://classroom.google.com/u/1/c/NjlxODU0MjkyNzMz>
 - qzjnsld
-
- Verilog Overview
 - Demo of EDA tools

VDF: Evaluation Criteria

Type of Evaluation	Contribution ECE313	Contribution ECE513
Weekly Assignment	25 %	15%
Project (RTL to GDS)	Not needed	20%
Mid-sem	30 %	25%
End-sem	45 %	40%



VDF: Assignments



- Individual Assignment
 - 4-5 assignments will be given (N-1 will be considered)
 - All assignments will be scaled to 10 Marks
 - After scaling, the assignment which has the lowest marks will be eliminated
 - Total obtained marks will be scaled appropriately to make the contribution of Assignments to 25% or 15%
-
- Assignment will be uploaded on Google Classroom
 - Typically one week time to complete assignment on Google Classroom
 - Do not request for extension, ideally you should submit well before deadline
 - **Strict Plagiarism check will be done [Beware of this !!! No excuse will be entertained]**

VDF: Project

S. No.	Assignment Topic	Week No. [Tentative]
1	Submission-1 [Logic Synthesis]	2-8
2	Submission-2 [Physical Design]	9-12



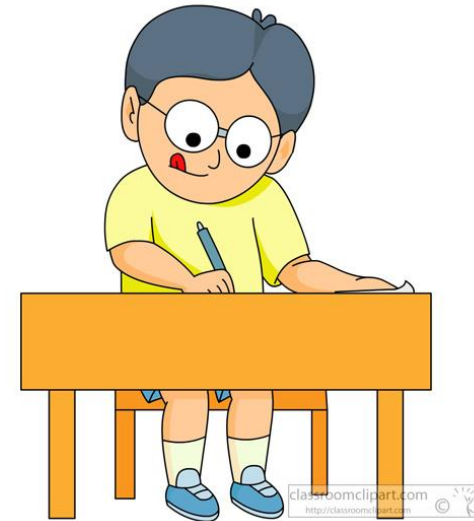
- Will be done in groups of 4 (Form a group on your own)
- Viva may be taken and different group members can get different marks
 - It will be the major component of marks in project
 - Will involve complete flow: RTL to GDS

VDF: Project

- You can choose the functionality to implement
 - You need to first submit the specification and get it approved (by second week)
 - You may change the specification slightly later on (after approval from TA)
 - The functionality should be non-trivial (it should have at least 500 logic gates after synthesis)
 - Two groups should not have the same functionality
-
- Do not download Verilog code from internet and use directly
 - You can look at the codes from internet and learn from them, without copying directly
 - Marks will be awarded based on originality of design, its complexity, analysis of the obtained result, and thoroughness of the work
 - **Plagiarism check will be done. Anything directly downloaded from internet (including GitHub) will be considered plagiarism.**

Mid-sem/End-sem Exams

- Questions will be asked from Assignment also
- Closed book



VDF: EDA Server

- Tools need to be run on EDA Server: UNIX
- Login for all the students will be created on this server
- Demo must be kept ready in your account
- You may be asked to give Demo in certain cases



References

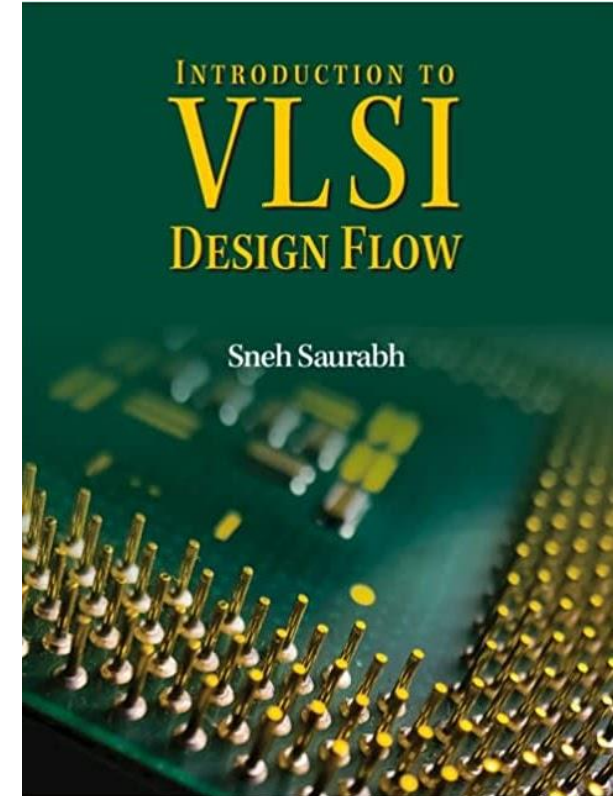
- References will be provided during lectures
- Saurabh, S. (2023). Introduction to VLSI Design Flow. Cambridge: Cambridge University Press.

NPTEL lectures:

<https://sites.google.com/site/snehsaurabhhome/teaching/np-tel-vlsi-design-flow-rtl-to-gds?authuser=0>

Pre-requisites:

https://www.youtube.com/playlist?list=PLKSq9uq3vZfKWLAQcC3WG_59rqwmz1zdW



Policies

VDF: Attendance Policy

- Student with less than 30% attendance [less than 8 classes] will be awarded FAIL grade *at the discretion of the instructor* [even when the total marks is above the PASS grade]
- Do not mark proxy attendance

ATTENTION: DO NOT Register for the course if you do not plan to attend the classes regularly (for ANY reason!)

VDF: Policies (1)

- Be punctual to class: up to 5 minutes late is tolerated, beyond that please do not come
- Cell phones, Laptops, IPODs, MP3, players or other portable electronic devices of any kind NOT allowed in classroom (should be switched off)



VDF: Policies (1)

- Be punctual in submission of Assignments and Projects
 - Each day of delay will attract 20 % penalty and after 5 days delay no more submission

Illustration: Day of submission is 20th August

- Submission is on-time if it is done by 12:00 midnight of 20th August
- Submitted at 8:00 am 21th August: Penalty of 20%, If marks obtained was 20, it will change to 16 marks
- Submitted at 1:00 am 25th August: Penalty of 100%, If marks obtained was 20, it will change to 0 marks



VDF: Plagiarism

- Academic dishonesty/plagiarism/cheating
 - IIITD rules will apply
 - Academic dishonesty policies of IIT Delhi apply.
<https://www.iiitd.ac.in/sites/default/files/docs/education/AcademicDishonesty.pdf>
 - Checks will be in place to detect copying/plagiarism/cheating



SSD: Questions/Doubts

1. Class: Many may have the same questions
2. Just After Class
3. Office Hours
4. During revision
5. No last minute query for assignment submission
 - Clear all your doubts at least 2 days before submission



VDF: Office Hours

Instructor:

12:30 pm - 1:30 pm: Wednesday, B-608

TAs

- Upal Majumder: 2:00-3:00 pm Monday
- Anubhav Kumar Singh: 2:30-3:30 pm Wednesday, Library Second Floor(casual study)
- Abhishek Chaurasiya

VDF: Expectations

- Revise last day lecture before coming
 - Revision will be done at the start of every lecture
-
- Ask questions and be active participants
 - Course expected to be at pace so that **EVERYONE** is able to follow
 - Give feedback

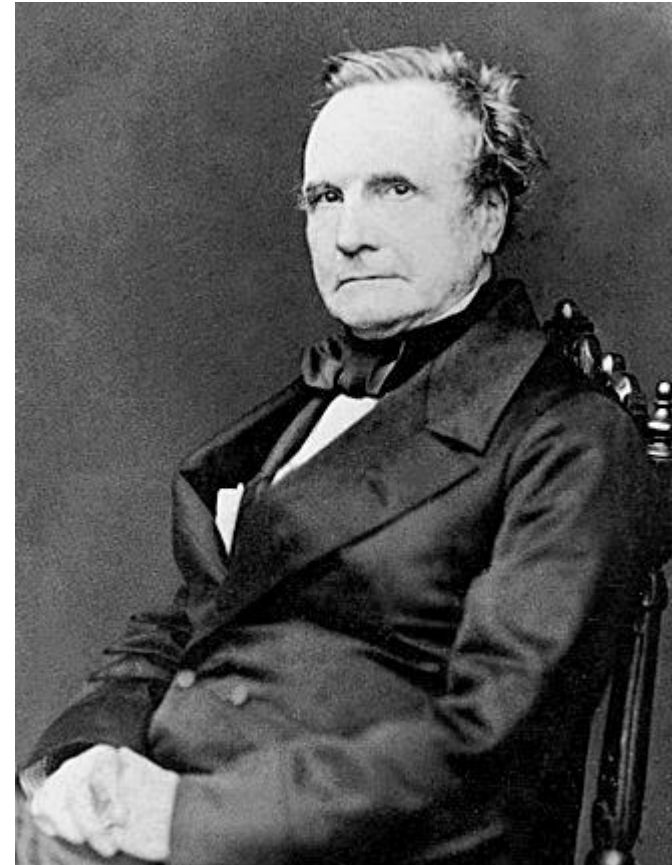


Introduction: Some Basic Concepts

VLSI: An Historical Perspective (1)

“...sources of excellence in the work produced by machinery depend on a principle which pervades a very large portion of all manufactures, and is one upon which the cheapness of the articles produced seems greatly to depend. The principle alluded to is that of **COPYING**, taken in its most extensive sense.”

—Charles Babbage, *On the Economy of Machinery and Manufactures*, Chapter 11, 1832



VLSI: An Historical Perspective (2)

VLSI

- Very Large-Scale Integration
- Integration
- Very Large Scale

Evolution of VLSI

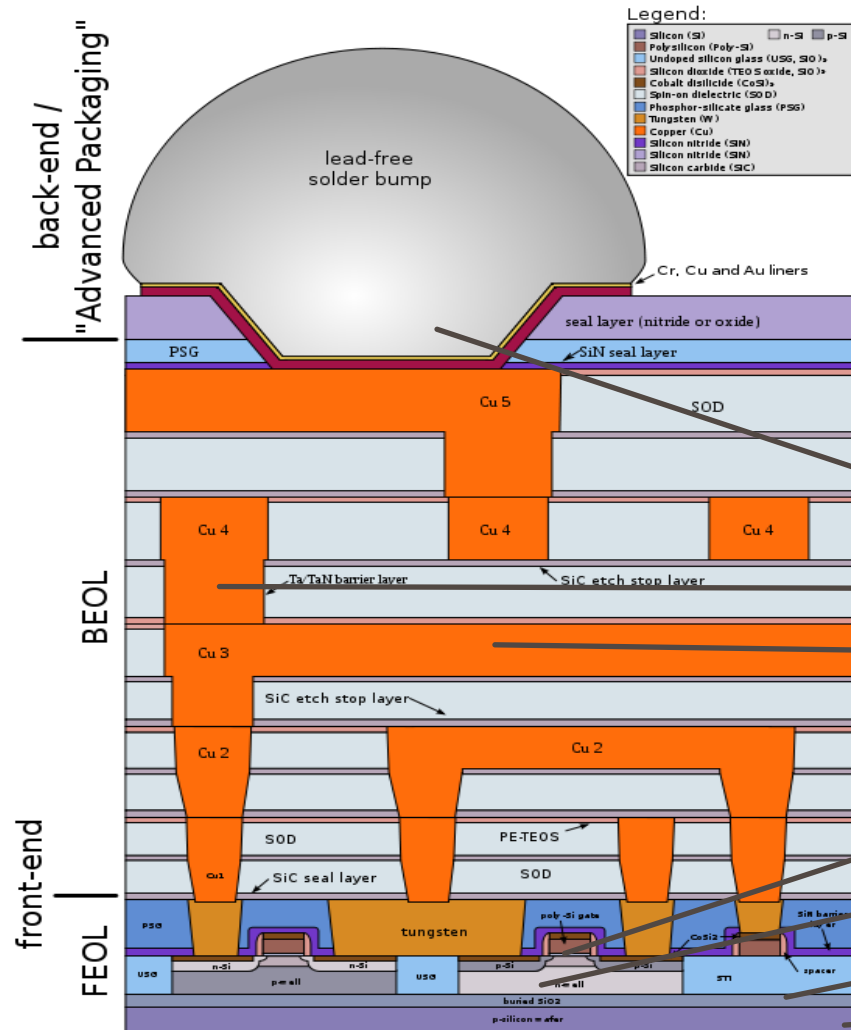
- More components being “crammed” into the integrated circuits (ICs)
- Moore’s prediction

1965 GORDON MOORE
IMAGINED A WORLD OF COMPUTING



“Integrated circuits will lead to such wonders as home computers – or at least terminal connected to a central computer – automatic controls for automobiles and personal portable communication equipment. The electronic watch only needs a display to be feasible today.”

Structure of Integrated Circuit (1)



- ICs are composed of multiple layers
- Diffusion layer, implant layer, metal layer etc.

- **Via** is used to make electrical connection between different layers and runs through two or more layers

For External Connection

Via
Metal
Routing

Transistor

Silicon

Silicon dioxide (Buried oxide)

Silicon Substrate

Figure shows a schematic of an IC built on Silicon on Insulator (SOI) wafer and having five metal layers

Source: Wikipedia

Structure of Integrated Circuit (2)

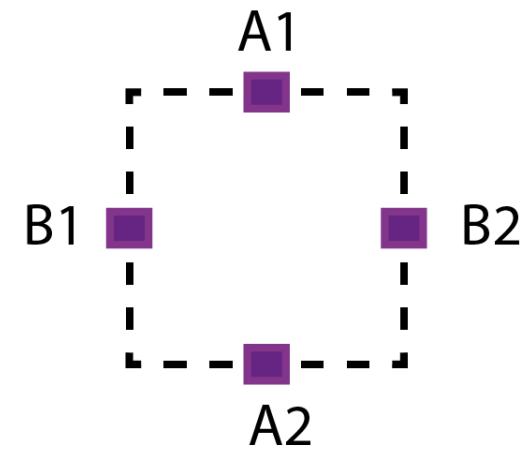
Problem 1:

Connect points A1 to A2 and B1 to B2.

B1/B2 should not be connected to A1/A2.

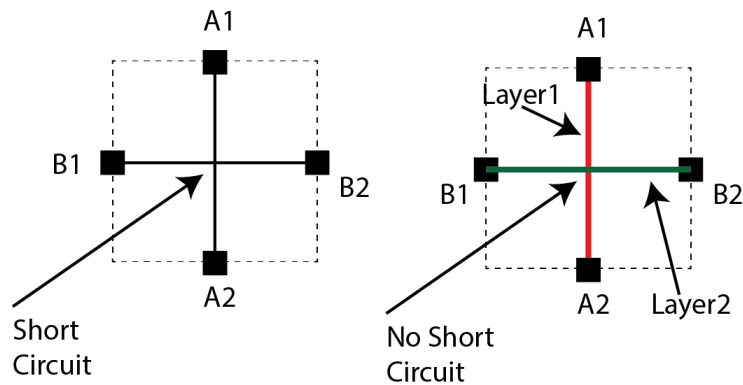
Connection can be made using wires, but wires cannot go outside the rectangle shown.

The wires are constrained to be in the plane containing A1, A2, B1, B2.

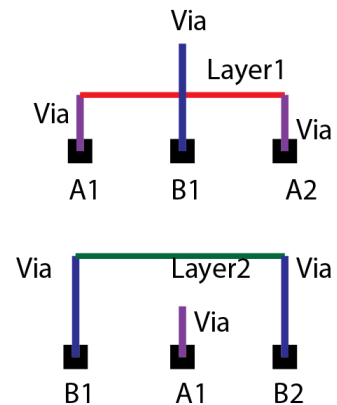


What happens when the constraint of wires being in the same plane is removed?

Structure of Integrated Circuit (3)

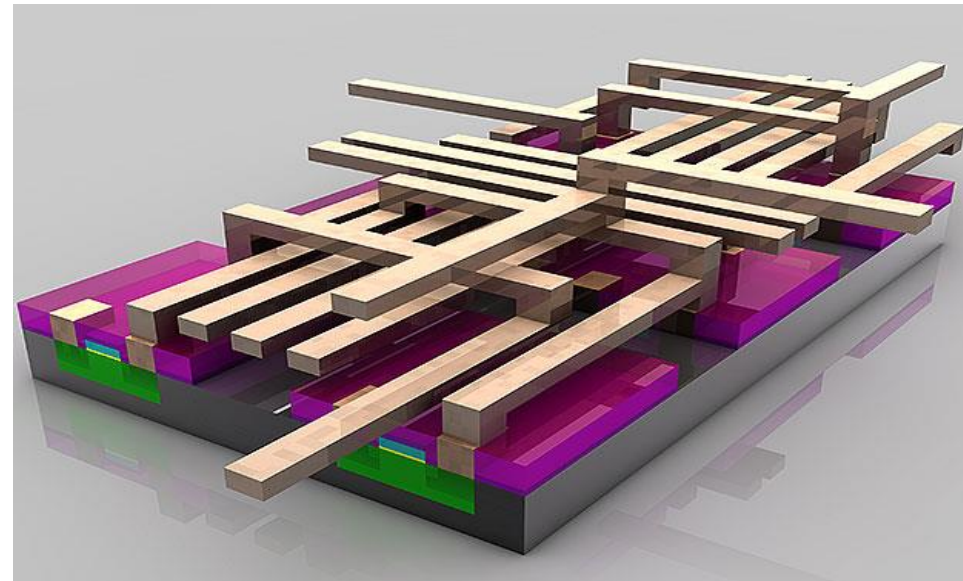


Connecting in One Layer



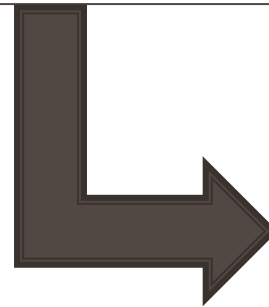
Connecting in Two Layers

- ✓ Multiple layers are necessary to make connections between devices that would otherwise short when connected in a single layer
- ✓ Layers defined by *mask* and fabricated using *photolithography*



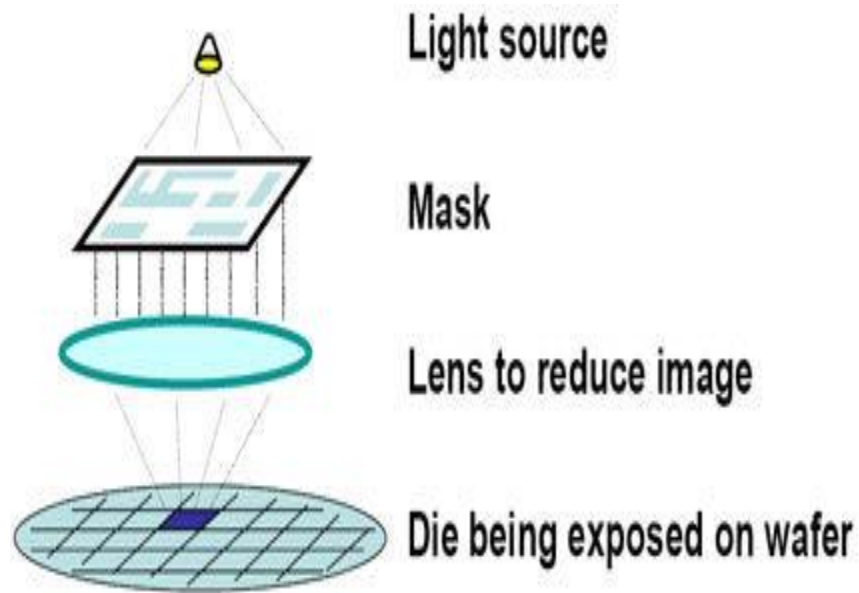
Source: <http://apcmag.com/picture-gallery-how-a-chip-is-made.htm/>

Introduction: Some Basic Concepts



Photolithography

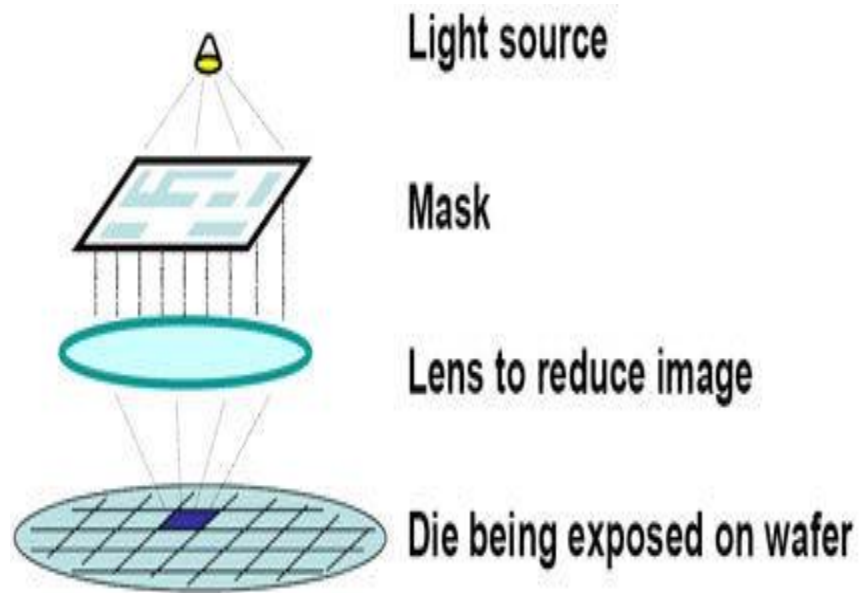
Photolithography (1)



- Key step in the fabrication of integrated circuits
- Process of transferring geometric shapes that are defined on a *mask* to the surface of a silicon wafer
- Steps similar to photography

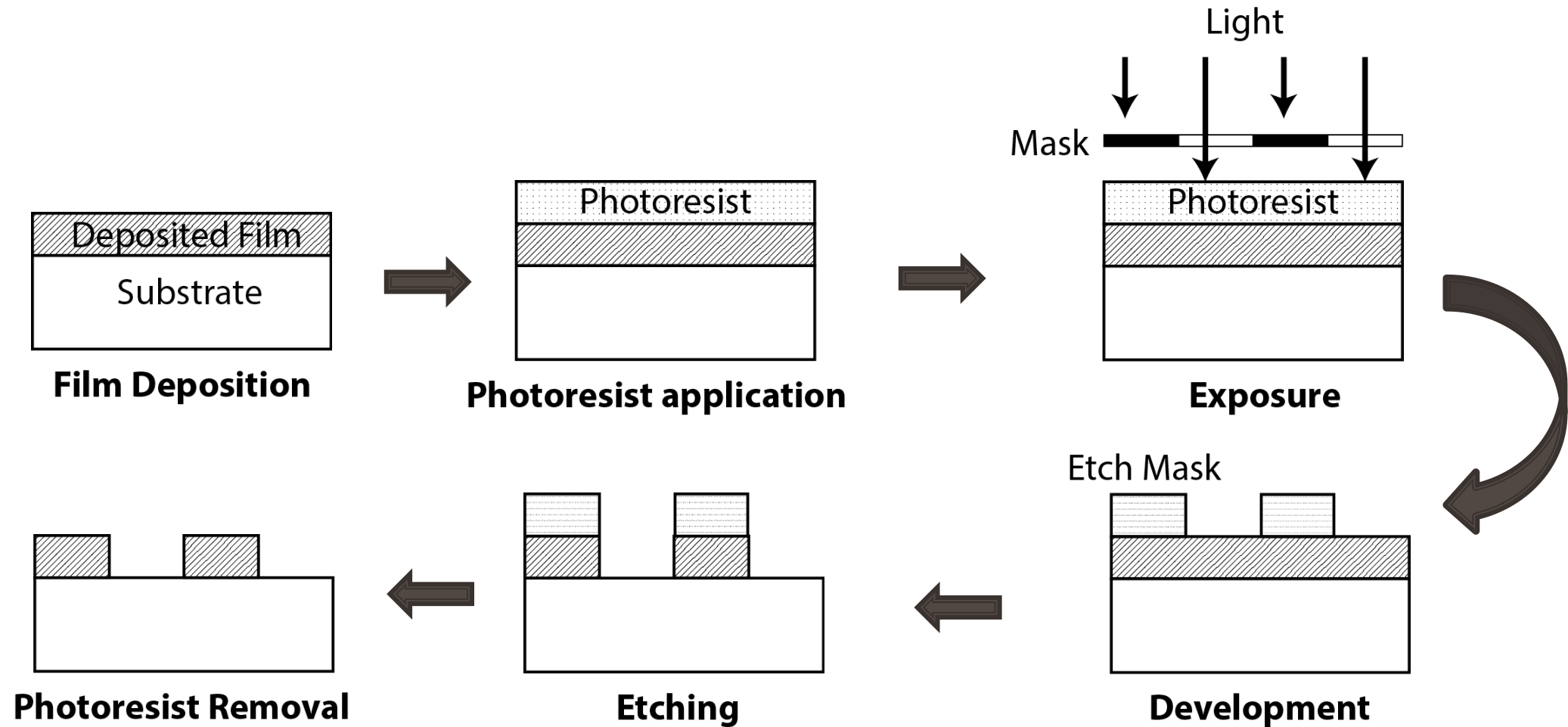
- Important components:
 - **Photoresist:** chemical sensitive to light
 - **Mask:** Defines the pattern that needs to be printed/transferred
 - **Light source and exposure system**

Photolithography (2)



- Exposure done in small area of wafer and **step/repeat** to cover complete wafer: ensures higher resolution
- For different layers of integrated circuits different masks are used and photolithography is carried out for each layer

Photolithography (3)



References

- S. Saurabh, “Introduction to VLSI Design Flow”. Cambridge: *Cambridge University Press*, 2023.